

# Tina Giorgadze

Email: [tg2876@columbia.edu](mailto:tg2876@columbia.edu) | LinkedIn: [TinaG](#) | Website: [tinagiorgadze.com](http://tinagiorgadze.com)

---

## EDUCATION

### **Columbia University, New York, NY**

Ph.D. candidate in Statistics

M.A. in Statistics, 2023-2025

M.Phil. in Statistics, 2026

### **Bard College, Annandale-on-Hudson, NY**

B.A. in Mathematics & Computer Science, May 2023 - **GPA 3.98**

- Distinguished Scientist Scholarship Recipient at Bard College, 2019-2023
- Dean's List Spring 2019 - Spring 2022

## SKILLS

**Programming:** R, Python, Java, MATLAB, LaTeX

**Machine Learning:** PyTorch, Autoencoders, Classification

**Statistics:** Bayesian Inference, Simulation-Based Inference, Likelihood-Free Inference, Monte Carlo Methods, Approximate Bayesian Computation (ABC), GLMs

**Data Tools:** Pandas, Excel, STATA

**Languages:** English (fluent), Georgian (native), Russian (upper-intermediate)

## PUBLICATIONS

- [Giorgadze T](#), Fischel H, Tessier A, Norton K-A. Investigating Two Modes of Cancer-Associated Antigen Heterogeneity in an Agent-Based Model of Chimeric Antigen Receptor T-Cell Therapy. *Cells*. 2022; 11(19):3165. <https://doi.org/10.3390/cells11193165>
- H. Fischel, [T. Giorgadze](#), A. Tessier and K. -A. Norton, "Computational Modeling of Chimeric Antigen Receptor (CAR) T-Cell Therapy of a Binary Model of Antigen Receptors in Breast Cancer," *2021 IEEE International Conference on Bioinformatics and Biomedicine (BIBM)*, 2021, pp. 3267-3274, DOI: 10.1109/BIBM52615.2021.9669393
- **Giorgadze T**, Tavaré S. "New Uses for Old Stochastic Processes: Models for DNA Pile-Ups from Single-Cell DNA Sequencing," in *Poisson Distribution - Methods, Models and Case Studies*, ISBN 978-1-80632-156-8. Accepted, forthcoming

## CONFERENCES/TALKS/PRESENTATIONS

- "Text Simplification Using Graph Algorithms", Women in Mathematics in New England, September 2022
- "Using a Computational Model to Simulate Chimeric Antigen Receptor (CAR) T-Cell Therapy in Triple-Negative Breast Cancer with Binary Distribution of Antigen Receptors", New York Celebration of Women in Computing, April 2022
- "Building an agent-based model to simulate breast cancer immunotherapy," 2021 IEEE International Conference on Bioinformatics and Biomedicine, BIBM 2021, December 2021

## RESEARCH EXPERIENCE

### **Columbia University, Irvine Institute of Cancer Dynamics - PhD Research Assistant (September 2023–present)**

- Develop likelihood-free statistical inference methods for inferring copy number variation from single-cell DNA sequencing data using simulation-based Bayesian approaches
- Design and implement simulation pipelines, statistical models, and machine learning algorithms in R to evaluate inference methods on synthetic and real genomic datasets
- Present research findings at weekly group meetings, help mentor summer research interns, and co-author a forthcoming book chapter on Poisson-process models for DNA sequencing

### **Bard Summer Research Institute in Machine Learning (BSRI) 2022 - Research Assistant (June 2022–July 2022)**

- Implemented a text simplification algorithm aimed at using neural networks and a sentence fusion graph to generate simplifications of an original sentence, and utilized machine learning tools to rank the generated sentences
- Implemented a sentence fusion graph from scratch using Python, extracting linguistic features from existing data, and training binary classification models to rank the sentences generated by my sentence fusion graph; our lexical simplification pipeline achieved up to 89% potential and 33% recall, combining three candidate generators

### **Bard Summer Research Institute in Computational Biology (BSRI) 2021 - Research Assistant (June 2021–May 2023)**

- Conducted computational biology research focused on building a 3D agent-based computational model of Triple-Negative Breast Cancer to simulate Chimeric Antigen Receptor (CAR) T-cell treatment

- Wrote code for the CAR T-cell immunotherapy module in MATLAB, developed 3D plots to visualize our model in MATLAB, conducted literature reviews, compiled introduction, methods, and results sections of the research paper; the paper was published in BIBM 2021 Proceedings

### **TEACHING (TA EXPERIENCE)**

- Introduction to Statistics Recitation Leader, *Spring 2026*
- Probability and Statistics for Data Science Recitation Leader, *Fall 2025*
- Introduction to Statistics with Calculus Recitation Leader, *Summer 2025*
- Introduction to Probability and Statistical Inference Recitation, *Spring 2025*
- Linear Regression Models Recitation, *Fall 2024*
- Introduction to Statistical Reasoning, *Fall 2023, Spring 2024*
- Across these courses (10-180 students), led recitations of 10-50 students, held office hours, graded homework and exams, wrote solutions, and helped develop homework problems

### **WORK EXPERIENCE**

#### **Bard College, Annandale-On-Hudson, NY**

##### ***Data Science Program Participant at Data Science Corps: Wrangle, Analyze, Visualize Program (September 2022–May 2023)***

- Collaborated with a local environmental organization to analyze operational data, identify trends, and support data-driven decision-making
- Partnered with a multidisciplinary team to collect, organize, and communicate analytical findings through reports and visualizations

##### ***Math & Computer Science Tutor (February 2020–May 2023)***

- Tutored 50+ middle school, high school, and college students in mathematics, computer science, and standardized test preparation (AMC, SAT, SHSAT, ISEE) through one-on-one instruction, office hours, and enrichment programs
- Held weekly office hours for Precalculus, Calculus I & II, Proofs, introductory programming, and data structures while supporting coursework, programming assignments, and independent projects
- Designed lesson plans and curricula for middle-school mathematics programs and provided individualized academic mentoring

##### ***Bard Mathematics Department's Hiring Committee Student Member (September 2021–December 2021)***

- Selected to serve on the Bard Mathematics Department hiring committee to hire a tenured professor
- Read 30 applications, provided feedback, participated in discussions, and met with selected shortlisted applicants

##### ***Creative and Analytical Math (CAMP) Program Teaching Assistant (August 2021)***

- Served as a teaching assistant at a virtual program CAMP. Hosted 50 middle and high school students, offering daily classes and workshops in math, computer science, and arts
- Assisted instructors, communicated with students and parents, ran a daily hour of online activities for students, hosted Zoom calls, and managed break-out sessions

### **AWARDS/ACHIEVEMENTS**

- Kenneth Bush '36 Memorial Scholar in Mathematics: a scholarship given annually to a junior who has demonstrated excellence in mathematics, 2022
- Serota Award for Computer Science recipient: awarded annually to an undergraduate in computer science who has shown promise and dedication in using technology to improve the human condition and make a positive impact on society, 2022

### **EXTRACURRICULAR EXPERIENCE**

- **Social Chair, Columbia University Department of Statistics.** Organize departmental social events for Ph.D. students in collaboration with faculty and administrators, including happy hours, game nights, holiday celebrations, and movie nights. (2024–present)
- **Founder and President of Bard Math Club.** Organized weekly problem-solving sessions and mathematics competitions for approximately 10 students. Managed club operations, including budgeting, outreach, speaker recruitment, and event planning. (Fall 2020–Spring 2023)
- **Upstate New York Math Team Coach.** Work with middle and high school groups on a monthly basis to prepare them for olympiads and competitions. (January 2021–January 2022)
- **Coordinator of Math and Girls Plus Inspiration Equals Success ([MAGPIES](#)) program.** Helped plan and organize monthly virtual and in-person math events for young girls